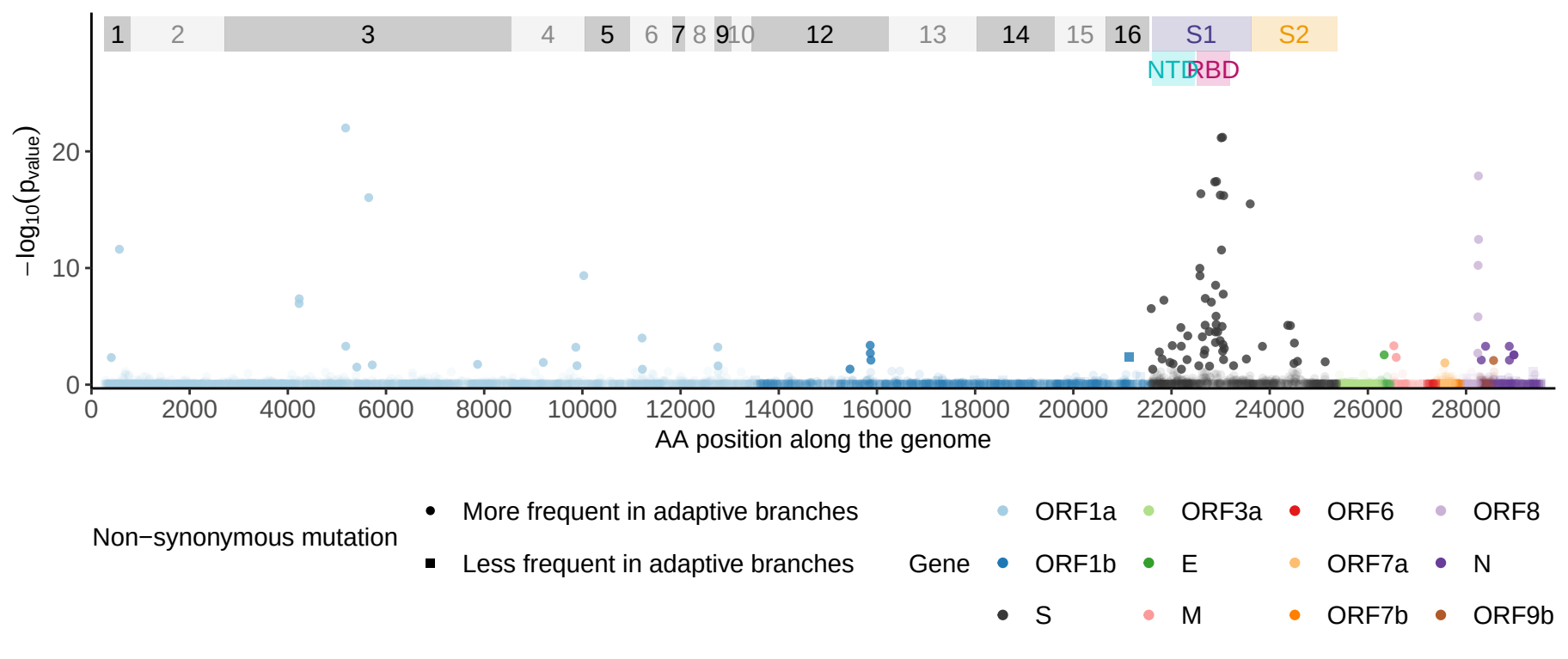

1
2
3
4
5
6
7
8
9
10
12
13
14
15
16
S1
S2
NTD
RBD
−log10(pvalue)
20
10
0
0
2000
4000
6000
8000
10000
12000
14000
16000
18000
20000
22000
24000
26000
28000
AA position along the genome
Non−synonymous mutation
More frequent in adaptive branches
Less frequent in adaptive branches
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF6
ORF7a
ORF7b
ORF8
N
ORF9b